

Shayan Shakeri

(416) 471-6651 | shayansnezhad@gmail.com | github.com/sshakerinezhad | merlyn-labs.com | Toronto, Canada

PROFESSIONAL SUMMARY

AI research engineer working across LLM and agent evaluation, RL environments, and VLA post-training. At BMO's AI Centre of Excellence: uncovered systematic risk-downplaying bias in a GenAI tool serving \$200B+ AUM, built the deterministic eval pipeline that catches it at scale, and built a multi-agent data-layer harness over bank client data. Co-founder of Merlyn Labs, researching VLAs and designing games for agent evaluations; M.Eng in AI & Robotics, University of Toronto, expected April 2027.

CORE COMPETENCIES

LLM & Agent Evaluation

RL Environments & Reward Design

VLA Post-Training (RL, Imitation Learning, Finetuning)

Failure Analysis

PyTorch, JAX

OmniGibson, MuJoCo

WORK EXPERIENCE

BMO AI Centre of Excellence Sep 2025 - Present

AI Research Engineer

Toronto

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving \$200B+ AUM wealth management
- Built a deterministic agent eval pipeline using hundreds of synthesized inputs to detect misaligned outputs at scale
- Built a graph-based agentic system from the ground up to answer multi-hop relational queries across bank client data; since grown into a multi-agent data-layer harness
- One prompt found, sorted, and screened nearly 30,000 clients for defined-benefit plans autonomously in 3 hours, previously requiring days of manual work
- Developing RL environments to train specialized agents for BMO's wealth management division
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy

Merlyn Labs Aug 2025 - Present

AI Researcher & Co-Founder

Toronto

- Showed $\pi 0.5$ generalization failures on LIBERO-PRO are recipe-induced overfitting; proposed an alternative baseline
- Conservative finetuning doubled position-swap success (21% to 42%), showing spatial generalization over memorization
- Developing VLM judges that score rollouts into dense, context-dependent RL rewards that are difficult to game
- Open-sourced flow-matching VLA integration for RLinf, enabling RL training on the BEHAVIOR-1K suite in OmniGibson
- Running sim-to-real transfer of household tasks on hand-built AlohaMini embodiment

Epineuron Technologies May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op

Mississauga

- Helped develop PeriPulse, an FDA Breakthrough-designated device now in multi-national clinical trials
- Designed and assembled PCBs and authored validation protocols (IEC 60601-1, ISO 13485) for clinical test runs
- Optimized power draw via oscilloscope/power analyzer testing, achieving 900% battery life improvement

RESEARCH & PROJECTS

Stanford BEHAVIOR-1K Challenge, 8th Place, Standard Track Sep - Nov 2025

Identified proprioceptive collapse as a critical VLA failure mode; masking 60% of proprioception improved task success by up to 48%. Found chunked execution outperformed temporal ensembling by 3x. Doubled manipulation success on long-tail subtasks by oversampling skill transitions via boundary resampling. Trained on 10,000+ demonstrations covering 22 of 50 scored task types; published a technical report (merlyn-labs.com/behavior-report) and further analysis on LessWrong.

EDUCATION

M.Eng in AI & Robotics University of Toronto Sep 2024 - Apr 2027, expected

B.Eng, Engineering Physics Co-op, Dean's Honour List McMaster University Sep 2018 - Jun 2023

SKILLS & INTERESTS

ML & AI: Python, C/C++, PyTorch, JAX, RL, Imitation Learning, LLM Fine-tuning, Alignment Evaluation, Agents
Robotics & Hardware: VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2, PCB Design, 3D Printing
Interests: Literature, Dungeons & Dragons, Volleyball, Twilight Imperium, Piano